

0:00	Title	“Good morning. My name is Martin Maxa, from the Czech Technical University in Prague, Department of Measurement. I am presenting joint work with Doctor Jakub Svatoš.” ■ detection, classification and localization of impulsive sounds — in practice, gunshots ■ “how we detect them, how we tell them apart, and how we find where they came from”
0:30	Why	“Shootings in public spaces are becoming more frequent, and gun ownership is rising worldwide.” ■ demand for autonomous security — campuses, hospitals, parks, government buildings ■ cameras, drones, guards give <i>some</i> protection — fall short with an active shooter ■ a camera has to be pointed the right way; sound does not care about line of sight
1:15	Signature	“So what does a gunshot look like to a microphone? This is a real recording — and the two components do not arrive together.” ■ shock wave first, ≈ 2.5 ms — only if $v > c$ — bullet drags a cone of compressed air across the mic — the N shape ■ [CLICK] muzzle blast ≈ 48 ms, <i>45 ms later</i> — gases leaving the barrel, 1–3 ms, travels only at the speed of sound ■ between and after = reflections — “remember that, it comes back at the end”
2:30	Naive	“You might think this is easy — just look at the spectrum, or just look at the waveform. Neither works on its own.” ■ spectrum alone → describes the environment, not the gunshot ■ time domain → distorted by nonlinear dispersion and obstacles ■ ⇒ feature extraction must combine both domains
3:00	System	“Here is the system. Each sensor is a stand-alone unit.” ■ listens continuously → sends the detected signal to a remote server ■ server does two things: classify the event, and find where it was, in 2D ■ why several units? coverage — and you need <i>two bearings</i> for a position
4:00	Localize	“So how do we actually find the shooter?” ■ two MEMS mics per unit, $d = 0.186$ m, $f_s = 44.1$ kHz — delay is at most ≈ 24 samples, <i>but that is enough</i> ■ $\Theta = \arcsin(c \Delta n / (d f_s))$ — speed of sound $\times$ delay, over spacing $\times$ sampling rate ■ [CLICK] one unit = a direction; two units cross = the position ■ measured: cross-correlation + parabolic interpolation, MAE 0.77°, 70 events, -45° to $+60^\circ$
5:00	Features	“Now the classification. The feature extractor is based on cepstral coefficients.” ■ same chain for all four: frame → FFT → power → log → DCT ■ what differs is the filter bank, and it is not on the diagram ■ MFCC (from speech, validated on ranges) LFCC linear IMFCC inverted GTCC gammatone = models the ear, more noise-resistant ■ tested: 4 extractors $\times$ 3 frame lengths (15/30/50 ms) $\times$ 2 classifiers
6:15	TABLE	“This is the result table for the nine-millimetre class. Let me walk you through it.” ■ rows grouped by frame length, four extractors inside each columns = ACC, RLC, MCC — each for SVM and NN ■ “now look only at the red rows” — GTCC is the best row in every single block ■ best cell: GTCC + NN at 50 ms → 99.15 / 96.77 / 0.98 ■ context: test set is 118 events, 31 of them 9 mm ⇒ 117 of 118 correct, recall 30 of 31
7:45	Numbers	“Three things come out of this.” ■ MFCC, LFCC, IMFCC almost identical — reshaping a mel-type bank buys you little ■ GTCC consistently better — gammatone resolves the low frequencies finely, and that is where muzzle-blast energy sits ■ NN beats SVM, and the gap grows with longer frames ■ longer frames do help, <i>but</i> 50 ms starts to include the room — the reflections from slide 3
8:45	Conclusion	“To summarise.” ■ distributed network of stand-alone sensors + server → detect, classify, localize in 2D ■ cepstral coefficients are an effective extractor, with both classifiers ■ GTCC beats the mel family; NN slightly better, especially at longer frames ■ validated with real firearms, several calibers and ammunition types
9:30		“Thank you — I am happy to take questions.”

If asked, the numbers. Corpus 371 train / 118 test, five classes (false alarms 86/21, **9 mm 70/31**, 5.56 NATO SD 100/31, 7.62 Tokarev 55/17, .22 60/18) | 26 features per extractor, $\approx 75/25$ split | NN: MATLAB Neural Fitting, Levenberg–Marquardt, 370 iterations | bearing: cross-corr 0.84°, +parabolic 0.77°, PBDE 3.24°, PBDE+linreg 3.91° | backup slides **11** setup **12** bearing **13** other cepstral **14** metrics (RLC = recall).

Running late? Three sentences are marked **[CUT IF LATE]** in the notes — **Why** (camera pointed the right way), **Signature** (cone of compressed air), **Numbers** (why gammatone wins). All three ≈ 35 s, and dropping them breaks nothing. **Checkpoint**: start the table at **6:15**; past 6:45, cut all three.

Do not guess: 2D position error in metres, SNR limits of *this* classifier, firing distances in this dataset. “We did not characterize that in this study — Jakub Svatoš ran the campaign, I can put you in touch.”